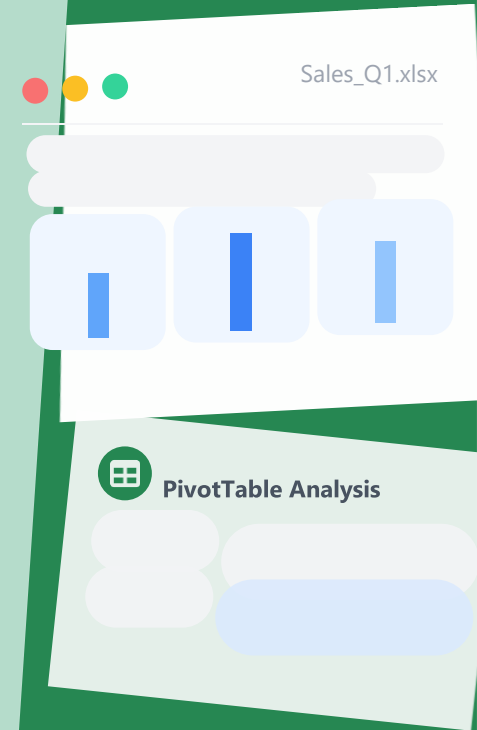




Microsoft Excel Advanced Level

 Instructor
Goh Kheng Yee



TRAINER PROFILE



Trainer Kheng Yee

- HRDC Accredited Trainer
- M.Sc. in Analytics, Georgia Tech, USA
- 6 years of Industrial Experience in AI & Data Science
- Trainings focused on AI, Machine Learning, Python, SQL, Data Analytics, and Data Visualization.

Professional Experience Highlights

Consulting

Led development of AI-powered learner engagement systems for an education institution. Contributed to public sector projects by applying advanced analytical models.

Media

Built AI/machine learning models to identify audience segments, enabling the shift to addressable TV advertising.

Banking

Developed machine learning models for banking products and optimized lead quality through model overlay techniques.

Professional Certificates



HRDC Accredited
Trainer



AWS Certified
Cloud Practitioner



Salesforce Certified
Tableau Desktop
Foundations



Google Cloud
Certified Cloud
Digital Leader

Module 1: Advanced Formula Techniques

</> Key Formula Techniques

INDEX + MATCH

Flexible lookups with row/column control

```
=INDEX (A:A, MATCH (B1, C:C, 0) )
```

OFFSET

Dynamic range referencing

```
=OFFSET (A1, 0, 0, 10, 1)
```

INDIRECT

Text-to-reference conversion

```
=INDIRECT ("Sheet"&A1)
```

Dynamic Arrays

Modern Excel functions

```
=FILTER (A:A, B:B>100)
```

💡 Practical Applications

- ✓ Replace VLOOKUP with INDEX+MATCH for left lookups across multiple columns
- ✓ Build dynamic named ranges with OFFSET for rolling 12-month analysis
- ✓ Parameterize models using INDIRECT to switch sheets based on control cells

Module 2: PivotTables & Analytics

Advanced PivotTable Features

Calculated Fields

Create custom metrics and derived calculations

Grouping & Categories

Group dates, numbers, and text fields

Slicers & Timelines

Interactive filtering with visual controls

Multi-dimensional Analysis

Analyze data across multiple dimensions

Practical Applications

- ✓ Build YTD, QoQ growth, and running totals in PivotTables
- ✓ Segment analysis by region, product, and customer tier
- ✓ Compare scenarios using slicers to filter cohorts quickly

Best Practice

Ensure data cleanliness (no blanks/merged cells) and proper field types for optimal PivotTable performance.

PivotTable Workflow

Raw Data
Source Table

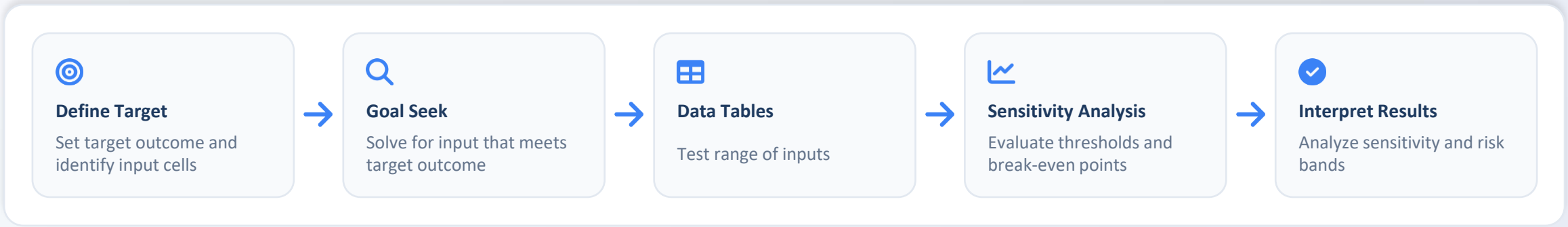


PivotTable
Analysis



Slicers
Filter

Module 3: What-If Analysis & Scenario Planning



Goal Seek Analysis

Target Value
\$50,000 (Profit)

Current Value
\$45,000

Input Cell
Price per unit

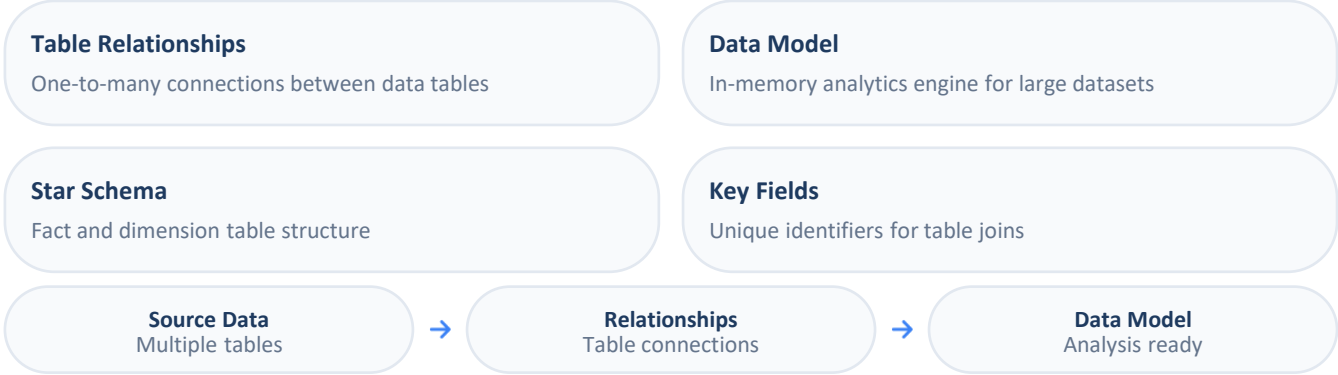
Required Change
+11.1%

Data Table Results

Scenario	Price	Volume	Revenue	Margin
Conservative	\$10	500	\$5,000	15%
Base Case	\$12	600	\$7,200	20%
Optimistic	\$15	800	\$12,000	25%

Module 4: Data Modelling & Relationships

Data Modelling Concepts



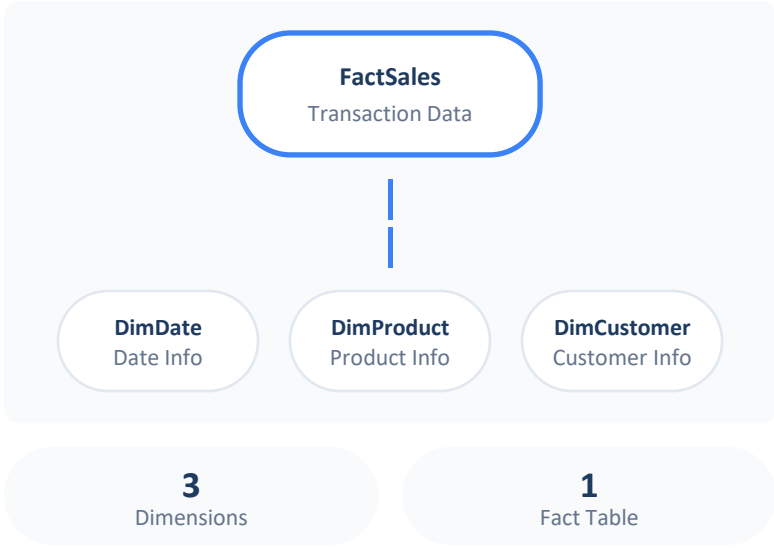
Practical Applications

- ✓ Build star schema to avoid repeated lookups and reduce file size
- ✓ Combine multiple data tables in a single Pivot without merges
- ✓ Enforce data integrity via relationships instead of nested formulas

Best Practice

Use consistent data types and unique keys for table relationships to ensure accurate joins.

Star Schema Visualization

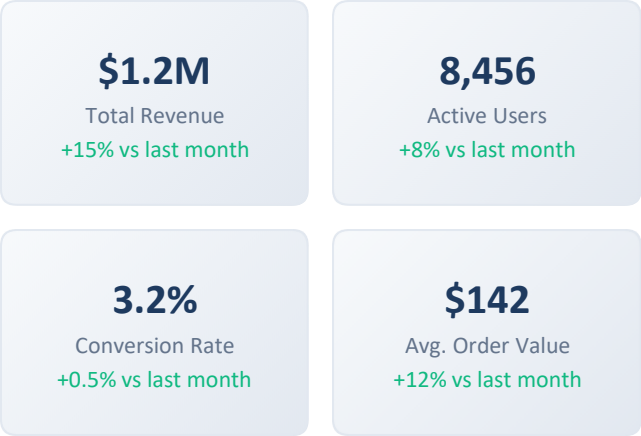


Key Benefits

- ✓ Scalable reporting without formula complexity
- ✓ Improved query performance with data model
- ✓ Reduced file size and memory usage

Module 5: Visualization & Dashboards

KPI Layout Examples



Color Usage Guidelines



Executive Summary Structure

- ✓ **Overview:** Start with key metrics and trends at the top
- ✓ **Context:** Provide background and comparison data
- ✓ **Insights:** Highlight actionable findings and recommendations
- ✓ **Next Steps:** Clear action items and owners

Chart Selection Guide



Design Do's & Don'ts

- | | |
|-------------------------|--------------------|
| ✓ Use consistent colors | ✗ Too many colors |
| ✓ Clear labels | ✗ Cluttered layout |
| ✓ Proper spacing | ✗ Small fonts |

Layout Principles

- Alignment: Use consistent grid system
- White Space: Allow breathing room
- Hierarchy: Visual weight and focus

Module 6: Macros & Automation

Recording Macros

- 1 Enable Developer Tab**
File → Options → Customize Ribbon → Check Developer
- 2 Start Recording**
Developer → Record Macro → Name & Description
- 3 Perform Actions**
Execute the tasks you want to automate
- 4 Stop Recording**
Click Stop Recording button
- 5 Test & Run**
Run macro from Macros dialog

Key Tips

-  Use relative references for dynamic ranges
-  Comment your code for clarity
-  Test macros on sample data first
-  Save as .xlsm for macro-enabled

Common Use Cases

[Report formatting](#) [Data cleaning](#) [Auto-refresh](#) [Batch processing](#)

Key Takeaways & Next Steps

✓ Module Summary



Module 1

Advanced formulas: INDEX+MATCH, OFFSET, INDIRECT



Module 2

PivotTables: calculated fields, slicers, grouping



Module 3

What-if analysis: Goal Seek, Data Tables



Module 4

Data modeling: relationships, star schema



Module 5

Dashboards & Visualization best practices



Module 6

Automation: macros, VBA basics

🏆 Key Achievements

- ✓ Mastered advanced formulas for flexible data analysis
- ✓ Built dynamic reports with modern Excel functions
- ✓ Created interactive dashboards with slicers
- ✓ Automated repetitive tasks with macros